



5/7/2026

# Food Trade in Egypt

## Analytics Graduation Project

 Final Project Report

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# 1. Executive Summary

## Food Trade in Egypt Analytics Project

DEPI R4 Program | May 2026

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### Project Overview

This project focuses on transforming raw international food trade data into actionable, decision-ready insights that support food security policies and strategic trade decision-making in Egypt. Through systematic data cleaning, analytical modeling, and interactive visualization, the project bridges the gap between complex trade datasets and clear, policy-relevant intelligence.

#### Dataset Size:

- 46,691 rows before data cleaning → 45,375 rows after data cleaning
- 378 food products, 178 partner countries, spanning 6 years (2018–2023)

### Key Objectives Achieved

| # | Objective                                                                           |
|---|-------------------------------------------------------------------------------------|
| 1 | Assess Egypt's food trade balance and export coverage ratio                         |
| 2 | Analyze trade partner concentration risk using the Herfindahl–Hirschman Index (HHI) |
| 3 | Identify top export products and strategically significant import items             |
| 4 | Measure dependency on imported grains, with a focus on wheat and maize              |
| 5 | Identify high-growth food products with strong investment potential                 |
| 6 | Compare export market diversification with import source concentration              |
| 7 | Forecast the export coverage ratio through 2027 under multiple scenarios            |
| 8 | Develop nine interactive dashboards across three analytical tools                   |

### Key Findings

- 1- Persistent food trade deficit of -\$53.5B (2018–2023).
- 2- Export coverage ratio improved from 36% to 49.6%.
- 3- High grain dependency (94% imported).
- 4- Import concentration risk (39.9% from top 3 countries).
- 5- Strong export diversification (178 partners, HHI = 0.033).

## Main Risks Identified

| Risk                                          | Level  | Impact                                  |
|-----------------------------------------------|--------|-----------------------------------------|
| Import concentration (3 countries = 39.9%)    | ● High | Geopolitical supply chain vulnerability |
| Meat production gap dependency (99% imported) | ● High | Food insecurity if global prices spike  |
| Grain dependency (94% imported)               | ● High | Structural production deficit           |

## Opportunities Identified

| Opportunity                 | Potential                                     |
|-----------------------------|-----------------------------------------------|
| Fruits & Vegetables exports | Competitive advantage (77%, 96% export share) |
| Chickens, Yoghurt, Eggs     | Early signals for VC & agri-investment        |
| Agro-processing             | Value-added export potential                  |

## Forecast

| Scenario    | Description                              | Coverage Ratio by 2027 |
|-------------|------------------------------------------|------------------------|
| Optimistic  | Agro-processing + import diversification | 62-65%                 |
| Base Case   | Continuation of current trend            | 58-62%                 |
| Pessimistic | Geopolitical or climate shock            | 50-52%                 |

## Strategic Recommendations

| # | Recommendation                                                                      | Priority     | Responsible Body        |
|---|-------------------------------------------------------------------------------------|--------------|-------------------------|
| 1 | Diversify grain import sources from 3 to 6+ countries (Kazakhstan, Romania, France) | ● Urgent     | Ministry of Supply      |
| 2 | Invest in agro-processing for fruits & vegetables                                   | ● High       | Ministry of Agriculture |
| 3 | Develop local meat and poultry production                                           | ● High       | Ministry of Agriculture |
| 4 | Establish strategic grain reserves (6+ months)                                      | ● Medium     | Ministry of Supply      |
| 5 | Create monthly monitoring system for key indicators                                 | ● Continuous | CAPMAS                  |

## Tools & Deliverables

| Tool     | Number of Dashboards |
|----------|----------------------|
| Excel    | 3                    |
| Tableau  | 4                    |
| Power BI | 2                    |
| Total    | 9 Dashboards         |

## Conclusion

Egypt's food trade shows improving resilience, as the export coverage ratio increased from 36% in 2018 to 49.6% in 2023. However, persistent vulnerabilities remain due to high import concentration, significant grain dependency, and structural gaps in meat production. Strengthening food security therefore requires targeted actions focusing on import diversification, agro-processing expansion, and enhanced local production to reduce long-term trade deficits.

## 2. Introduction

### 2.1 Project Idea & Data Source

This project analyzes Egypt's food trade during the period 2018–2023 using real, non-synthetic data obtained from the Food and Agriculture Organization (FAO) database. The dataset covers all food product categories and trading partners related to Egypt's import and export activities, providing a comprehensive basis for analysis.

Dataset Size:

| Metric               | Value         |
|----------------------|---------------|
| Rows before cleaning | 46,691        |
| Rows after cleaning  | 45,375        |
| Food products        | 378           |
| Partner countries    | 178           |
| Years                | 6 (2018-2023) |

### 2.2 Why This Topic?

| # | Reason                                           |
|---|--------------------------------------------------|
| 1 | Based on real (non-synthetic) data               |
| 2 | Directly focused on Egypt                        |
| 3 | Food security is a national priority             |
| 4 | Provides actionable insights for decision-makers |

### 2.3 Project Scope

| Dimension  | Scope                                 |
|------------|---------------------------------------|
| Geographic | Egypt (Reporter country: Egypt)       |
| Temporal   | 2018–2023 (six years)                 |
| Thematic   | All food products (imports & exports) |

### 3. Project Objectives

| # | Objective                                                                  |
|---|----------------------------------------------------------------------------|
| 1 | Assess Egypt's food trade balance and export coverage ratio                |
| 2 | Analyze key trade partners and concentration risks                         |
| 3 | Identify the most impactful products in exports and imports                |
| 4 | Measure dependency on imported grains (wheat and maize)                    |
| 5 | Identify promising export opportunities based on high-growth products      |
| 6 | Compare export market diversification with import source concentration     |
| 7 | Forecast the export coverage ratio until 2027                              |
| 8 | Build an interactive dashboard summarizing the complete food trade picture |

### 4. Key Questions

#### 4.1 Strategic Questions (7 Questions)

| #  | Question                                                                         |
|----|----------------------------------------------------------------------------------|
| Q1 | How is Egypt's food trade balance evolving, and is the coverage ratio improving? |
| Q2 | Who are the most important trade partners, and is there a concentration risk?    |
| Q3 | Which products drive exports, and which place the highest burden on imports?     |
| Q4 | How dependent is Egypt on imported grains (Wheat and Maize)?                     |
| Q5 | Which products show dramatic growth spikes as early signals for investment?      |
| Q6 | How diversified are export markets compared to import sources?                   |
| Q7 | What is the forecast for the export coverage ratio by 2027?                      |

## 4.2 Analytical Questions per Dashboard (9 Dashboards)

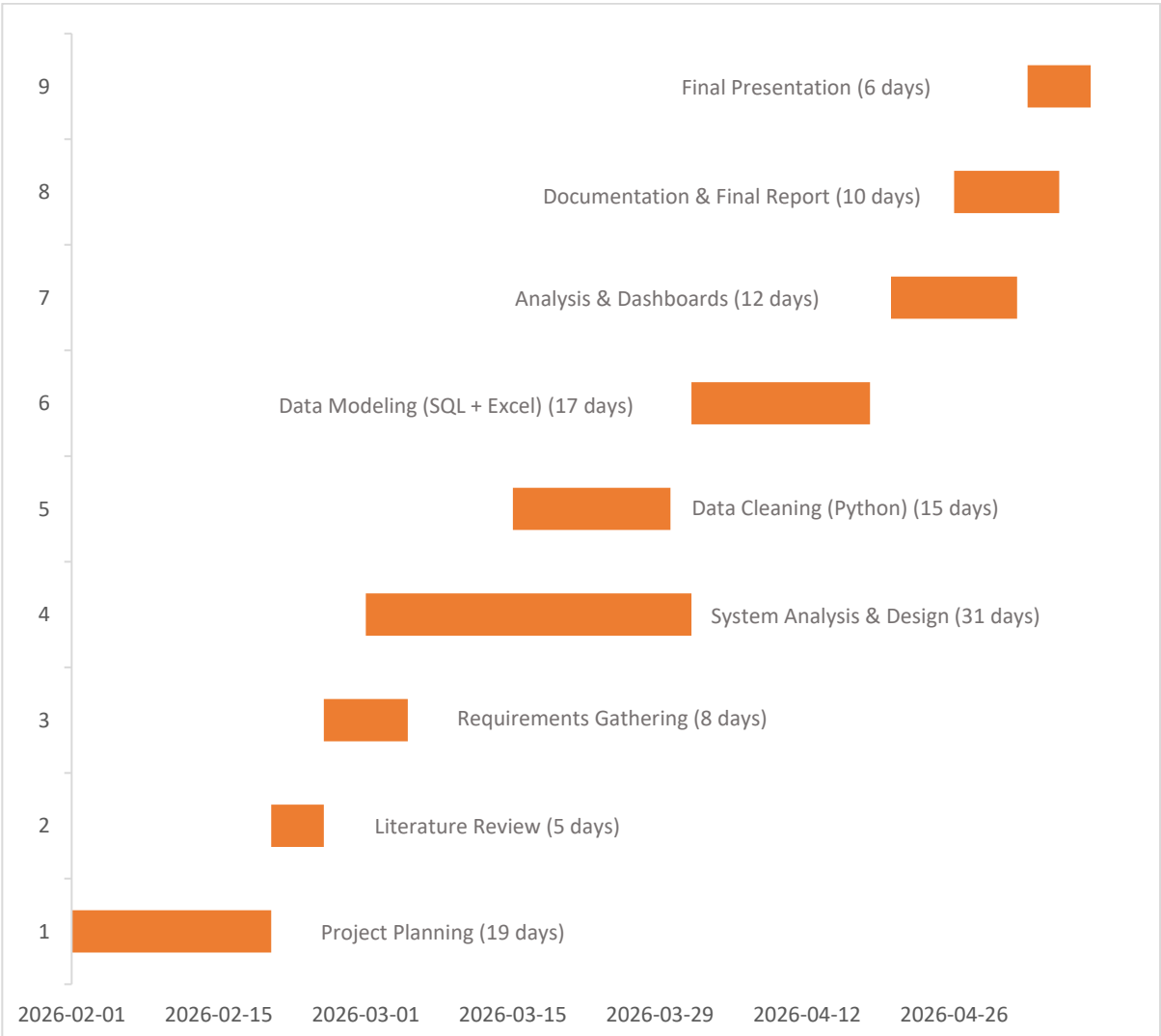
| # | Dashboard                       | Question                                                                     |
|---|---------------------------------|------------------------------------------------------------------------------|
| 1 | Trade Balance & Coverage        | Is there structural improvement in Egypt's trade balance and coverage ratio? |
| 2 | Categories: Risk vs Opportunity | Which categories represent export opportunities versus import risks?         |
| 3 | Top Products & Partners         | Which products and partners are the largest and fastest-growing?             |
| 4 | Partner Risk & Concentration    | Are import partners more concentrated and riskier than export partners?      |
| 5 | Growth & Opportunities Matrix   | Which products combine high growth with high stability?                      |
| 6 | Why Scenario Forecasting?       | Why is forecasting needed (shocks, deficits, and deviations)?                |
| 7 | Forecasting & Scenarios         | When can the coverage ratio reach 60%, and under which scenario?             |
| 8 | Categories Trends Over Time     | How does each category's share evolve over time?                             |
| 9 | Interactive Executive Dashboard | What is the complete picture, and where are risks concentrated?              |



## 5. Project Planning & Management

### 5.1 Timeline (Gantt Chart)

| Phase                        | Start     | End       | Duration |
|------------------------------|-----------|-----------|----------|
| Project Planning             | 1/2/2026  | 20/2/2026 | 20 days  |
| Literature Review            | 20/2/2026 | 25/2/2026 | 5 days   |
| Requirements Gathering       | 25/2/2026 | 5/3/2026  | 10 days  |
| System Analysis & Design     | 1/3/2026  | 1/4/2026  | 30 days  |
| Data Cleaning (Python)       | 15/3/2026 | 30/3/2026 | 15 days  |
| Data Modeling (SQL + Excel)  | 1/4/2026  | 18/4/2026 | 18 days  |
| Analysis & Dashboards        | 20/4/2026 | 2/5/2026  | 12 days  |
| Documentation & Final Report | 26/4/2026 | 6/5/2026  | 10 days  |
| Final Presentation           | 3/5/2026  | 9/5/2026  | 6 days   |



## 5.2 Task Assignment

| Phase                  | Tasks                                                   | Responsible Members |
|------------------------|---------------------------------------------------------|---------------------|
| Data Cleaning (Python) | Defining data cleaning strategy and rules               | All members         |
|                        | Python script execution and final implementation        |                     |
| Data Modeling (SQL)    | Database design, star schema, and ETL processes         | All members         |
| Data Modeling (Excel)  | Initial calculations and pivot tables                   | All members         |
| Questions & Insights   | Developing analytical questions and extracting insights | All members         |
| Dashboard Development  |                                                         |                     |
|                        | Dashboards 1, 2 (Excel)                                 | All members         |
|                        | Dashboard 3 (Excel)                                     |                     |
|                        | Dashboard 4 (Tableau)                                   |                     |
|                        | Dashboards 5, 6, 7 (Tableau)                            |                     |
|                        | Dashboards 8, 9 (Power BI)                              |                     |
| Documentation          | Report writing and formatting                           | All members         |
| Team Leadership        | Project management and coordination                     | Amina Adel Soliman  |

## 5.3 Risk Assessment & Mitigation Plan

| # | Risk                                 | Probability | Impact | Mitigation                                                                      |
|---|--------------------------------------|-------------|--------|---------------------------------------------------------------------------------|
| 1 | Large dataset size (46,691 rows)     | High        | Medium | Python was used instead of Excel to ensure efficient processing and scalability |
| 2 | Missing quantity values (3,809 rows) | High        | High   | Applied a price-ratio-based imputation algorithm to estimate missing quantities |
| 3 | Inconsistent measurement units       | Medium      | Medium | Standardized all units using Python during the data cleaning phase              |
| 4 | Tool incompatibility                 | Low         | Medium | Ensured version alignment across tools and conducted regular backups            |
| 5 | Time constraints                     | Medium      | High   | Developed a realistic project schedule with built-in time buffers               |

## 5.4 Key Performance Indicators (KPIs)

| # | KPI                            | Target | Achieved |
|---|--------------------------------|--------|----------|
| 1 | Number of dashboards developed | 6-9    | ✓ 9      |
| 2 | Imputation success rate        | >60%   | ✓ 65.5%  |
| 3 | Number of generated insights   | >30    | ✓ 40+    |
| 4 | Number of tools used           | 3      | ✓ 3      |
| 5 | Complete project documentation | 100%   | ✓        |

## 6. Literature Review

| Source            | Topic                     | Relevance                    |
|-------------------|---------------------------|------------------------------|
| FAO (2023)        | Food trade statistics     | Primary data source          |
| World Bank (2024) | Egypt Economic Monitor    | Data validation              |
| CAPMAS (2024)     | Egyptian trade statistics | Macroeconomic interpretation |

### Research Gap Filled by This Project:

No previous study combined:

This project addresses a clear research and practice gap by integrating several components that have not been jointly explored in previous studies. Specifically, no prior work combined large-scale data cleaning using Python, star schema modeling implemented through SQL Server, and multi-tool analytical workflows utilizing Excel, Power BI, and Tableau within a single coherent framework. In addition, this project uniquely delivers nine integrated dashboards designed to collectively answer seven strategic, policy-relevant questions, thereby bridging the gap between raw trade data, advanced analytics, and decision-ready insights.

## 7. Stakeholder Analysis

| Stakeholder             | Primary Interest                              | Needs from Project                                                  |
|-------------------------|-----------------------------------------------|---------------------------------------------------------------------|
| Ministry of Supply      | Food security and import policy design        | Recommendations for supplier diversification and risk reduction     |
| Ministry of Agriculture | Export development and market expansion       | Identification of high-potential export products and target markets |
| Investors Association   | Investment opportunities in agri-food sectors | High-growth, stable, and scalable products                          |
| Academic researchers    | Methodological rigor and reproducibility      | A transparent and replicable analytical framework                   |
| DEPI Jury               | Project quality and learning outcomes         | Compliance with graduation project requirements                     |

## 8. Requirements Gathering

### 8.1 Functional Requirements

| #    | Requirement                                                                                  |
|------|----------------------------------------------------------------------------------------------|
| FR-1 | The system can import and clean FAO trade data                                               |
| FR-2 | The system can calculate the Herfindahl–Hirschman Index (HHI) to assess market concentration |
| FR-3 | The system can compute the export coverage ratio                                             |
| FR-4 | The system can perform time-series trend analysis                                            |
| FR-5 | The system can forecast the export coverage ratio up to 2027                                 |
| FR-6 | The system can export analytical results to interactive dashboards                           |

### 8.2 Non-Functional Requirements

| #     | Requirement           | Specification                                                     |
|-------|-----------------------|-------------------------------------------------------------------|
| NFR-1 | Query performance     | System response time less than 5 seconds for standard queries     |
| NFR-2 | Data accuracy         | Error rate maintained below 1% after data cleaning and validation |
| NFR-3 | Scalability           | Ability to seamlessly incorporate additional years of data        |
| NFR-4 | Tool compatibility    | Full compatibility with Excel, Power BI, and Tableau              |
| NFR-5 | Documentation quality | Complete and well-structured code and project documentation       |

## 9. System Analysis & Design

### 9.1 Problem Statement

Raw FAO food trade data is unstructured and contains missing values, inconsistent measurement units, and varying data quality across countries and years, As a result, the dataset cannot be directly used for food security analysis or policy-oriented decision-making without systematic cleaning, transformation, and analytical modeling.

### 9.2 System Architecture & Data Flow

The system follows a structured data pipeline that transforms raw trade data into decision-ready analytical dashboards, The overall architecture is illustrated as follows:

**CSV Files (FAO) → Python (Cleaning) → MS SQL (Star Schema) → Dashboards (Excel/Power BI/Tableau)**

#### Pipeline Steps:

The analytical pipeline consists of the following stages:

- 1. Importing libraries**  
Core Python libraries including Pandas and NumPy were used for data manipulation and numerical processing.
- 2. Reading FAO data**  
Raw trade data was imported from FAO CSV/Excel files into the Python environment.
- 3. Exploratory data inspection**  
Initial exploration was conducted to examine dataset structure, dimensions, data types, and missing values.
- 4. Data cleaning and preprocessing**  
Missing values, inconsistent units, and invalid records were handled using rule-based and algorithmic cleaning methods.
- 5. Exploratory data analysis (EDA)**  
Descriptive statistics and trend exploration were performed to understand trade patterns and anomalies.
- 6. KPI calculation**  
Key performance indicators were computed, including the Herfindahl–Hirschman Index (HHI), export coverage ratio, and coefficient of variation (CV).
- 7. Database modeling**  
Cleaned and transformed data was exported to SQL Server and organized using a star schema design to support analytical queries.
- 8. Dashboard development**  
Nine interactive dashboards were developed across Excel, Power BI, and Tableau to answer seven strategic research questions.

### 9.3 Data Cleaning Examples (Python)

This section presents selected examples from the Python-based data cleaning process applied to the FAO dataset, highlighting key challenges and corresponding solutions

#### Example 1: Cleaning the Quantity Column

python: To ensure numeric consistency, the *Quantity* column was converted to numeric format, and invalid or non-positive values were treated as missing.

```
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
df.loc[df['Quantity'] <= 0, 'Quantity'] = np.nan
```

This step standardized quantity values and eliminated erroneous entries that could distort analytical results.

#### Example 2: Standardizing Measurement Units

python: FAO data contained quantities reported in inconsistent units, such as “1000 An”. These units were standardized to ensure comparability.

```
kilo_animal_mask = df['Quantity_Unit'] == '1000 An'
df.loc[kilo_animal_mask, 'Quantity'] = df.loc[kilo_animal_mask, 'Quantity'] * 1000
df.loc[kilo_animal_mask, 'Quantity_Unit'] = 'An'
```

This transformation ensured that all quantities were expressed using a unified base unit.

#### Example 3: Imputing Missing Quantities Using Price Ratios

python: Missing quantity values were estimated using a price-ratio imputation approach based on trade value and median unit prices.

Two hierarchical median levels were applied:

- Level 1: Category, Year, and Trade Flow
- Level 2: Year and Trade Flow (fallback level)

```
median_L1 = df.groupby(['Category', 'Year', 'Flow'])['Unit_Price'].transform('median')
median_L2 = df.groupby(['Year', 'Flow'])['Unit_Price'].transform('median')
final_median = median_L1.fillna(median_L2)
estimated_qty = np.where(
    (df['TradeValue_USD'] > 0) & (final_median > 0),
    (df['TradeValue_USD'] / final_median).round(2),
    np.nan)
df['Quantity'] = df['Quantity'].fillna(pd.Series(estimated_qty, index=df.index))
```

This approach maximized data retention while maintaining economic consistency.

## **Results:**

### IMPUTATION QUALITY REPORT

Total rows originally missing Quantity: 3,809

Successfully estimated (Price Ratio) : 2,493

Cannot impute (TradeValue also 0/null): 1,316

Imputation success rate : 65.5%

| Stage           | Row Count |
|-----------------|-----------|
| Before Cleaning | 46,691    |
| After Cleaning  | 45,375    |
| Dropped rows    | 1,316     |

Rows were dropped only when imputation was infeasible due to missing or zero trade values, ensuring data integrity without excessive loss.

## **9.4 Database Design (Star Schema)**

### **Dimension Tables:**

SQL

```
CREATE TABLE countries (  
    country_id INT IDENTITY PRIMARY KEY,  
    country_name VARCHAR(100) UNIQUE NOT NULL  
);
```

```
CREATE TABLE products (  
    product_id INT IDENTITY PRIMARY KEY,  
    item_code_fao INT NOT NULL,  
    product_name VARCHAR(255),  
    category VARCHAR(100)  
);
```

```
CREATE TABLE years (year INT PRIMARY KEY);  
CREATE TABLE trade_flows (flow_id INT IDENTITY PRIMARY KEY, flow_type VARCHAR(10));  
CREATE TABLE units (unit_id INT IDENTITY PRIMARY KEY, unit_name VARCHAR(20));  
CREATE TABLE quantity_sources (source_id INT IDENTITY PRIMARY KEY, source_type VARCHAR(20));
```

**Fact Table:**

```
CREATE TABLE trade_fact (  
    trade_id BIGINT IDENTITY PRIMARY KEY,  
    reporter_country_id INT NOT NULL,  
    partner_country_id INT NOT NULL,  
    product_id INT NOT NULL,  
    year INT NOT NULL,  
    flow_id INT NOT NULL,  
    unit_id INT NOT NULL,  
    source_id INT NOT NULL,  
    quantity DECIMAL(18,4),  
    trade_value_usd DECIMAL(18,2),  
    dataset_primary_key VARCHAR(255),  
    FOREIGN KEY (reporter_country_id) REFERENCES countries(country_id),  
    FOREIGN KEY (partner_country_id) REFERENCES countries(country_id),  
    FOREIGN KEY (product_id) REFERENCES products(product_id),  
    FOREIGN KEY (year) REFERENCES years(year),  
    FOREIGN KEY (flow_id) REFERENCES trade_flows(flow_id),  
    FOREIGN KEY (unit_id) REFERENCES units(unit_id),  
    FOREIGN KEY (source_id) REFERENCES quantity_sources(source_id)  
);
```



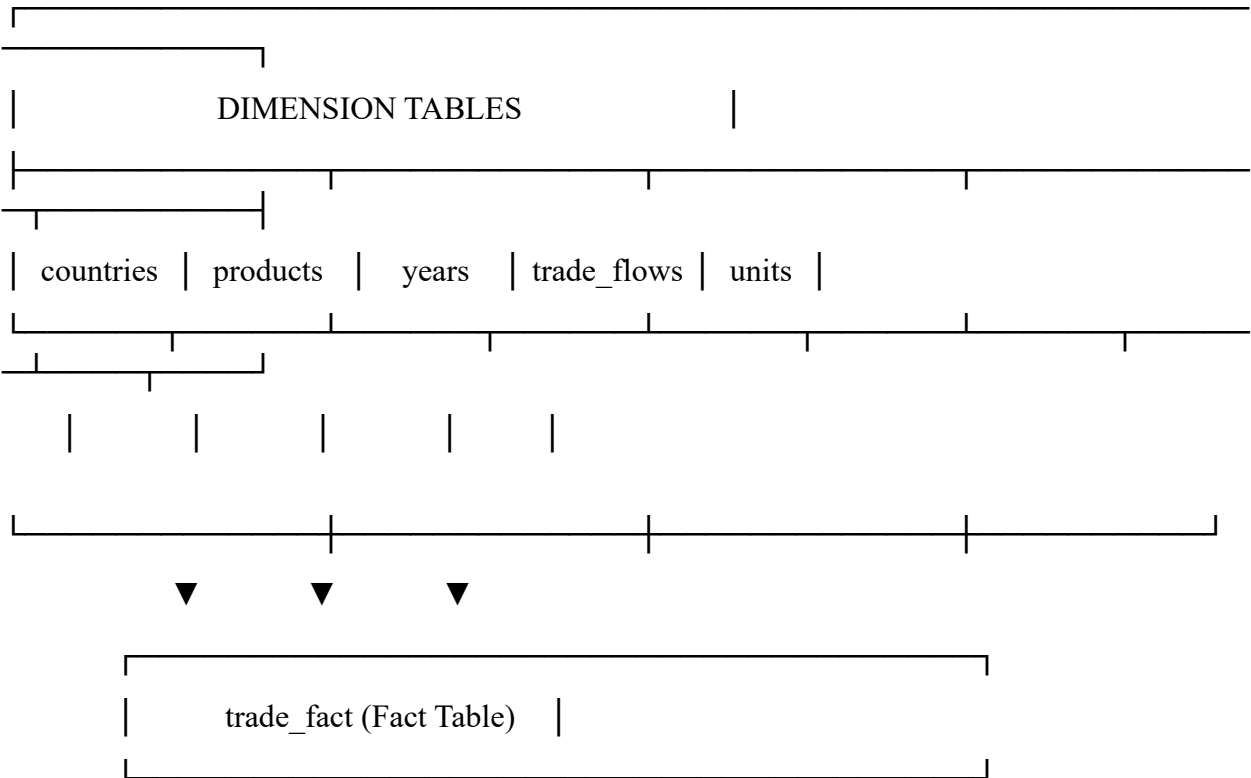
**Indexes for Performance:**

SQL

```
CREATE INDEX idx_fact_year ON trade_fact(year);
CREATE INDEX idx_fact_product ON trade_fact(product_id);
CREATE INDEX idx_fact_reporter ON trade_fact(reporter_country_id);
CREATE INDEX idx_fact_partner ON trade_fact(partner_country_id);
```

**Star Schema Diagram:**

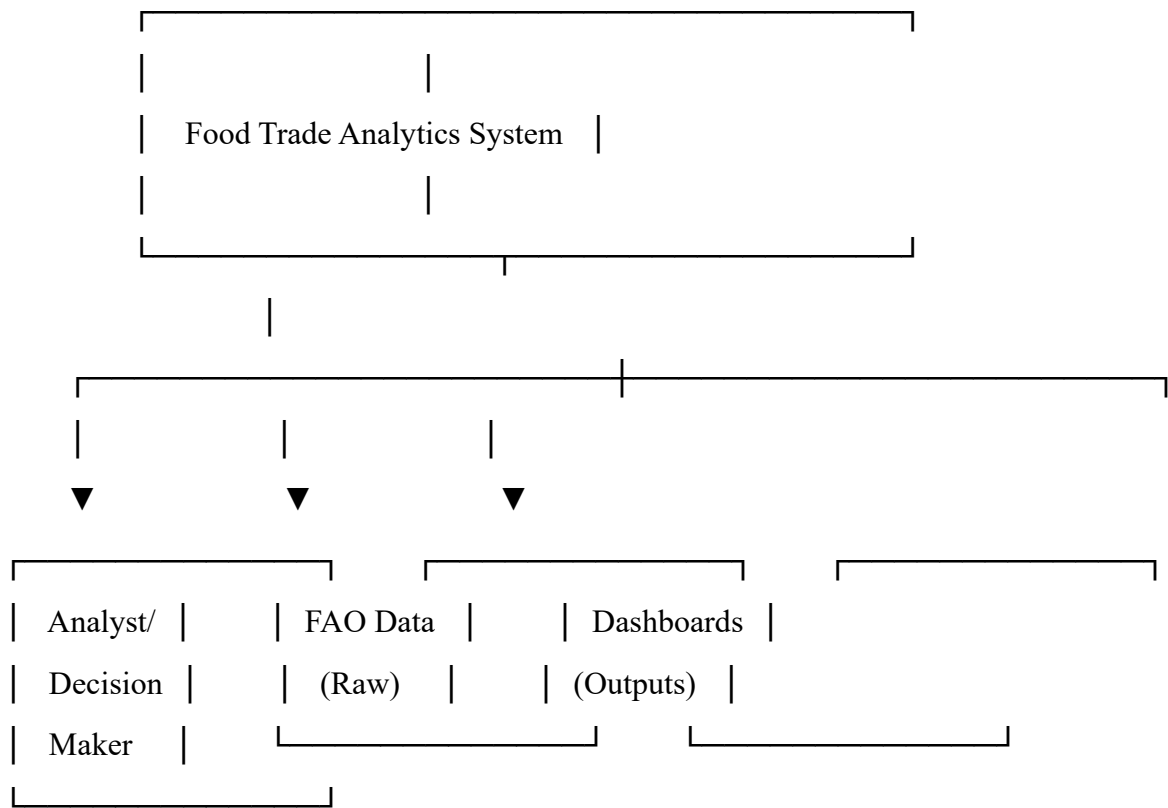
text



9.5 Data Flow Diagram (DFD)

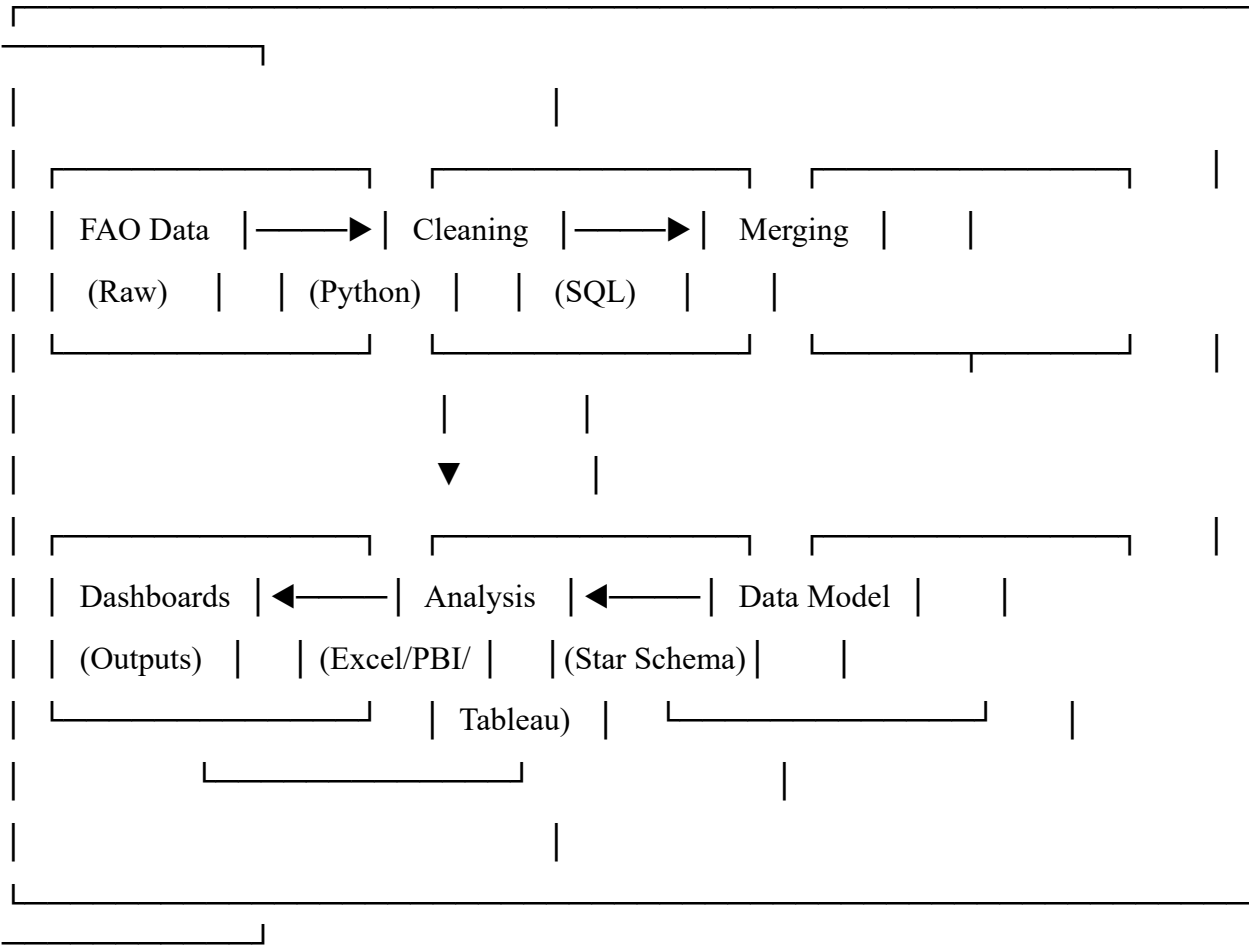
Context Level:

text



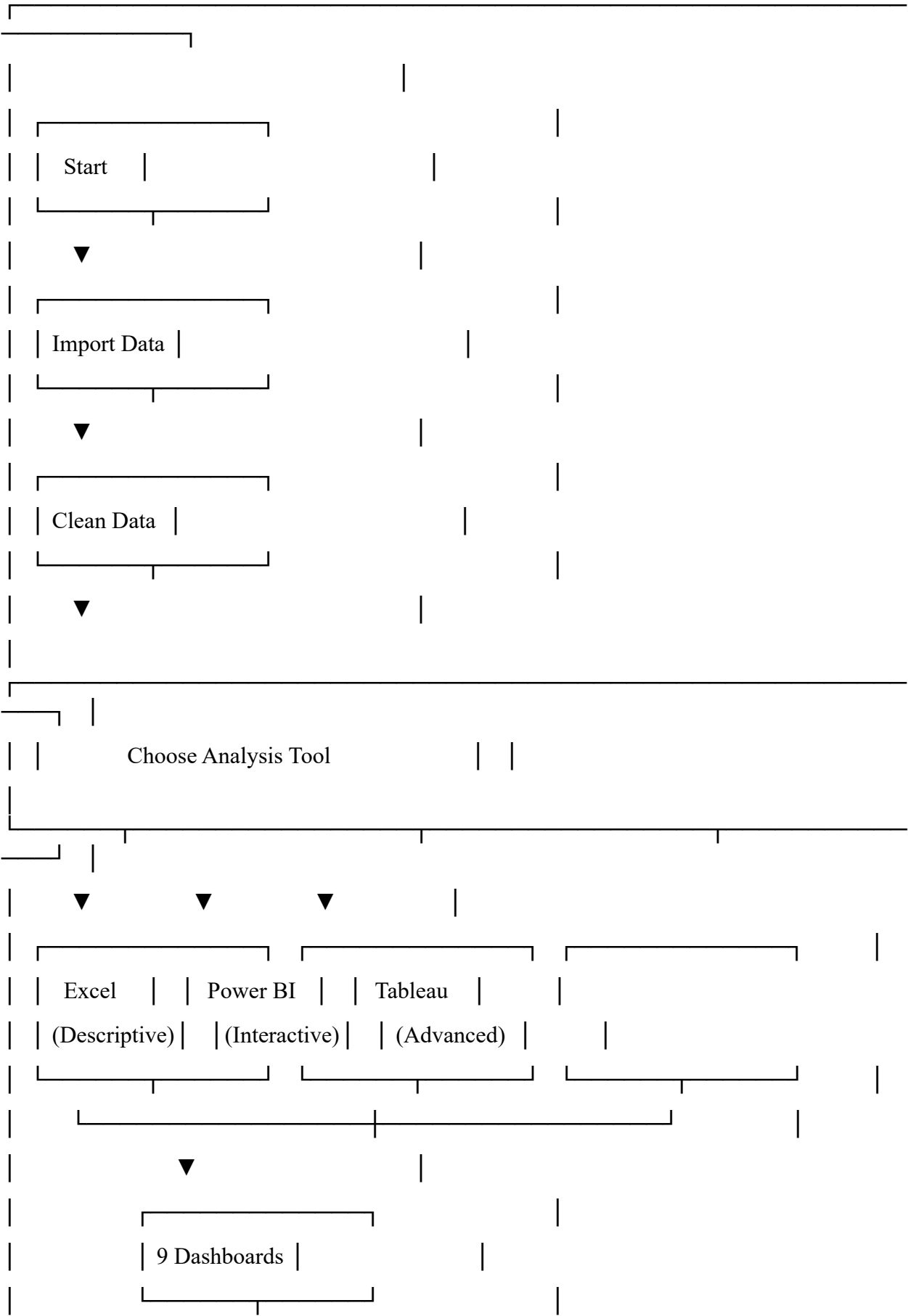
Level 1 DFD:

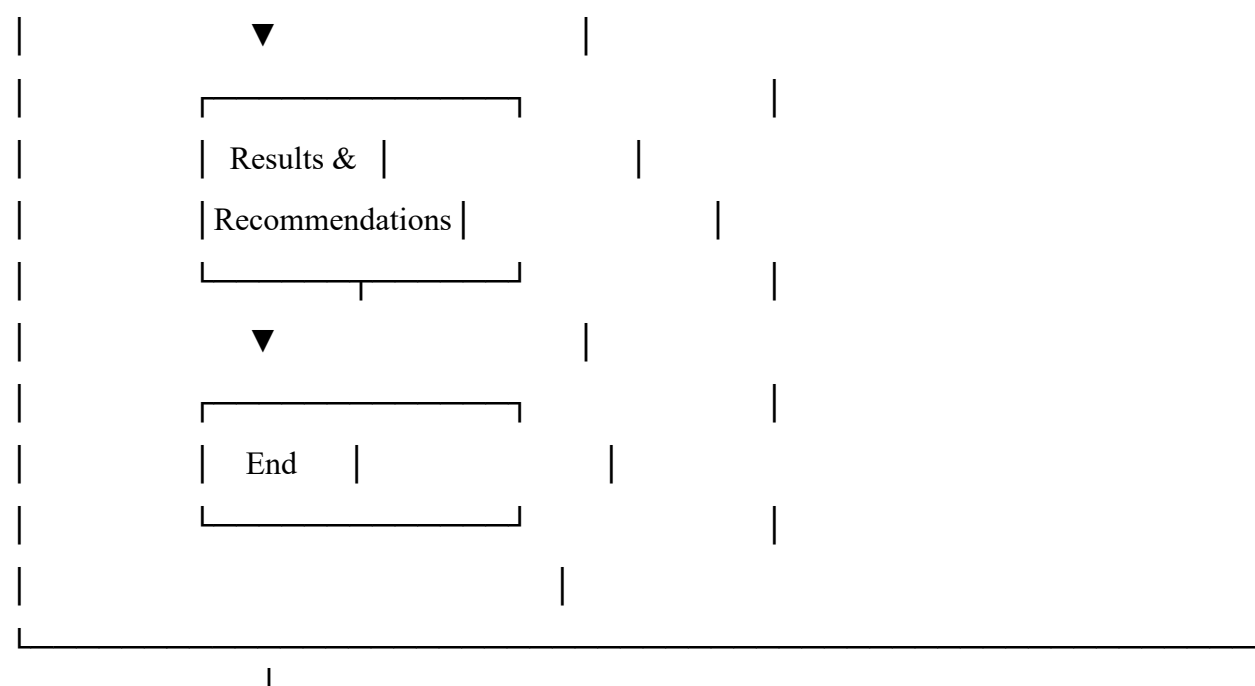
text



9.6 Activity Diagram (Analysis Process)

text





## 9.7 Technology Stack

| Layer            | Technology          | Version | Purpose                                           |
|------------------|---------------------|---------|---------------------------------------------------|
| Data Processing  | Python              | 3.9.7   | Data cleaning, preprocessing, and imputation      |
| Python Libraries | pandas, NumPy       | latest  | Data manipulation and numerical analysis          |
| Database         | SQL Server          | 2019    | Star schema implementation and analytical storage |
| Modelling        | Excel (Power Pivot) | 2019    | Initial calculations and data modeling            |
| Visualization    | Excel               | 2019    | Development of 3 descriptive dashboards           |
| Visualization    | Power BI            | Desktop | Development of 2 interactive dashboards           |
| Visualization    | Tableau             | 2026.1  | Development of 4 advanced analytical dashboards   |

## 10. Analytical Questions & Insights

This section synthesizes the key analytical insights derived from the developed dashboards. The findings focus on trade balance dynamics, import dependency risks, export diversification patterns, and indicators of structural improvement over time.

### Key Insights:

| # | Insight                                                                                                                                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Chronic food trade deficit</b> – Egypt exhibits a persistent food trade deficit amounting to approximately <b>USD 53.5 billion</b> over the period <b>2018–2023</b> , reflecting structural reliance on food imports.        |
| 2 | <b>Improving export coverage ratio</b> – Export coverage improved from <b>36% in 2018</b> to <b>49.6% in 2023</b> , indicating a gradual but incomplete narrowing of the trade gap.                                             |
| 3 | <b>High grain dependency risk</b> – Cereals are <b>94% import-dependent</b> , with wheat and maize imports totaling <b>USD 30.9 billion</b> , representing a significant food security vulnerability.                           |
| 4 | <b>Strong export market diversification</b> – Egypt exports food products to <b>178 partner countries</b> , with a low <b>HHI of 0.033</b> , reflecting excellent export market diversification and limited concentration risk. |

## Supporting Insights:

- Meat and poultry dependency – The meat and poultry category is 99.2% import-dependent, indicating a structural domestic production gap.
- Import concentration risk – Imports are moderately concentrated, with Russia, the United States, and Brazil accounting for 39.9% of total food imports.
- Export competitive advantage – Oranges represent Egypt's largest food export, totaling approximately USD 3.98 billion, highlighting a strong competitive position in the fruits category.
- High growth spikes – Exceptional growth rates were observed in chickens (+16,800%) and yogurt (+10,154%), serving as early signals of emerging investment opportunities (noting base-effect influence).
- Forecasting outlook – Under the base-case scenario, the export coverage ratio is projected to reach 58–62% by 2027, indicating continued but gradual improvement.

## 11. Dashboards (Analytical Story Sequence)

### The Story Flow:

**Current Situation (Q1, Q3, Q4) → Risk (Q2) → Opportunity (Q5) → Future (Q7) → Complete Picture**

### 11.1 Dashboard 1: Trade Balance & Coverage

| Element      | Detail                                                                                               |
|--------------|------------------------------------------------------------------------------------------------------|
| Tool         | Excel                                                                                                |
| Key Question | Is there structural improvement in Egypt's trade balance and export coverage ratio?                  |
| Chart Type   | Combo chart (Exports & Imports as bars, Coverage Ratio as line) with YoY growth bars                 |
| KPIs         | Total exports, total imports, trade balance, export coverage ratio, year-over-year growth indicators |

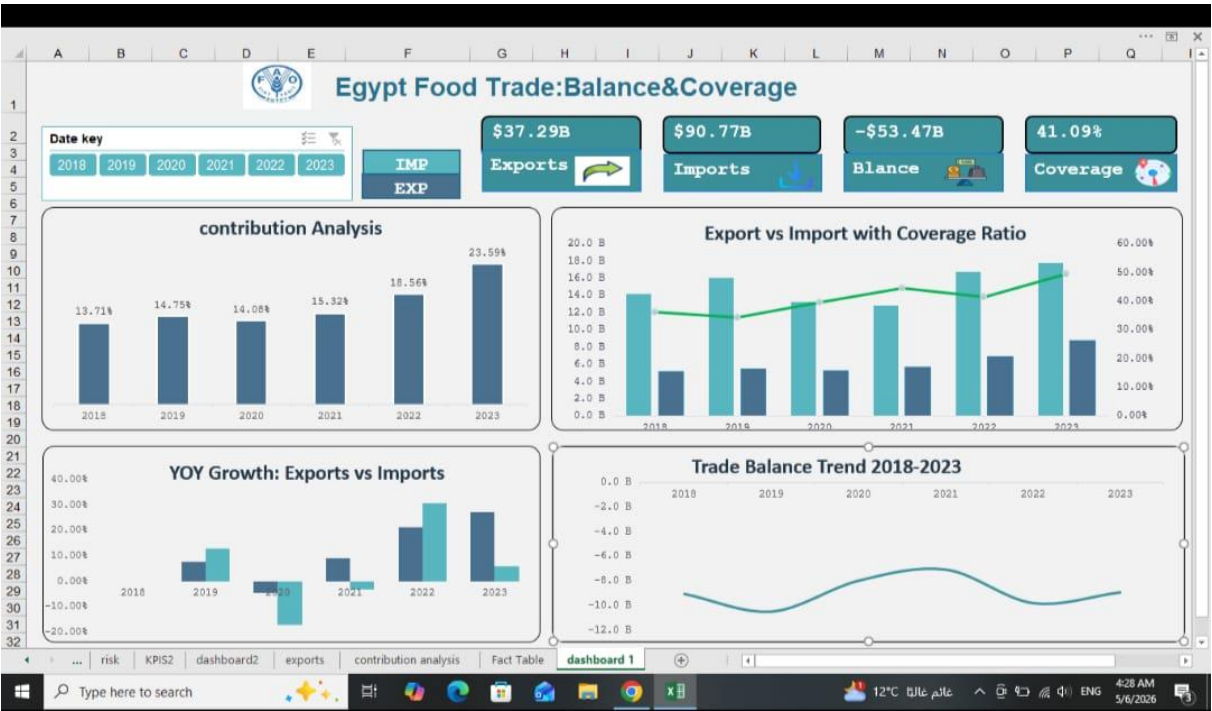
### Results:

| Year | Exports | Imports  | Balance   | Coverage |
|------|---------|----------|-----------|----------|
| 2018 | \$5.11B | \$14.19B | -\$9.08B  | 36.0%    |
| 2019 | \$5.50B | \$16.01B | -\$10.51B | 34.4%    |
| 2020 | \$5.25B | \$13.25B | -\$8.00B  | 39.6%    |
| 2021 | \$5.71B | \$12.82B | -\$7.11B  | 44.6%    |
| 2022 | \$6.92B | \$16.74B | -\$9.82B  | 41.3%    |
| 2023 | \$8.80B | \$17.75B | -\$8.95B  | 49.6%    |

Recommendations:

- 1. Continue supporting food exports (+27% growth in 2023)
- 2. Monitor external shocks (COVID 2020, Ukraine war 2022)

Figure 1: Trade Balance & Coverage Dashboard





## 11.2 Dashboard 2: Categories – Risk vs Opportunity

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| Element      | Detail                                                                 |
| Tool         | Excel                                                                  |
| Key Question | Which categories are export opportunities vs import risks?             |
| Chart Type   | 100% Stacked Bar + Grouped Bar                                         |
| KPIs         | Highest Export Category, Highest Import Category, Highest Dependency % |

### Results:

| Category       | Exports | Imports  | Dependency   | Direction |
|----------------|---------|----------|--------------|-----------|
| Fruits         | \$9.07B | \$2.71B  | 23% imported | ✅ Export  |
| Vegetables     | \$4.78B | \$0.19B  | 4% imported  | ✅ Export  |
| Processed Food | \$6.40B | \$5.29B  | 45% imported | Balanced  |
| Oils & Fats    | \$1.61B | \$8.68B  | 84% imported | ⚠️ Import |
| Cereals        | \$2.05B | \$33.14B | 94% imported | 🔴 Risk    |
| Meat           | \$0.07B | \$9.13B  | 99% imported | 🔴 Risk    |

### Top 3 Import Risk Categories:

| Category        | Dependency |
|-----------------|------------|
| Meat & Poultry  | 99.2%      |
| Cereal & Grains | 94.2%      |
| Live Animals    | 90.9%      |

### Recommendations:

1. Invest in local meat production (chicken, cattle)
2. Expand processed fruit exports (agro-processing)

Figure 2: Categories – Risk vs Opportunity Dashboard



### 11.3 Dashboard 3: Top Products & Partners

|              |                                                                                  |
|--------------|----------------------------------------------------------------------------------|
| Element      | Detail                                                                           |
| Tool         | Excel                                                                            |
| Key Question | Which are the largest and fastest-growing products & partners?                   |
| Chart Type   | Horizontal Bar Chart + Variance Bars + CAGR Line                                 |
| KPIs         | Largest Export (Value), Largest Export (Quantity), Highest Growth %, Top Partner |

#### Results – Top Products:

| Product | Value    | Rank               |
|---------|----------|--------------------|
| Oranges | \$3.98B  | 1 (largest export) |
| wheat   | \$18.11B | 1 (largest import) |

#### Results – Fastest Growing Products (CAGR):

| Product                     | CAGR (2018→2023) |
|-----------------------------|------------------|
| Brewing or distilling dregs | 261%             |
| Chickens                    | 179%             |
| Persimmons                  | 167%             |
| Yoghurt                     | 152%             |

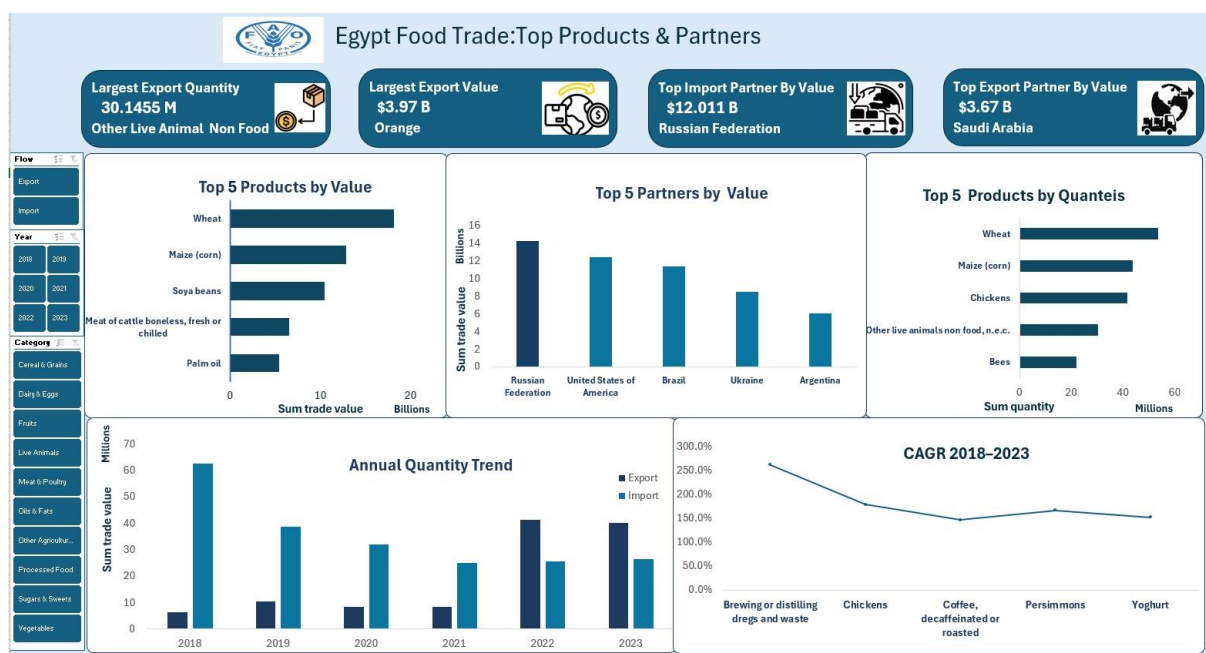
#### Results – Top Partners:

| Partner      | Role                  | Value    |
|--------------|-----------------------|----------|
| Russia       | Largest import source | \$12.01B |
| Saudi Arabia | Largest export market | \$3.68B  |

#### Recommendations:

1. Monitor Chickens and Yoghurt for sustainable growth (base effect warning)
2. Use these "Early Signals" for investment decisions

**Figure 3: Top Products & Partners Dashboard**



## 11.4 Dashboard 4: Partner Risk & Concentration

| Element      | Detail                                                         |
|--------------|----------------------------------------------------------------|
| Tool         | Tableau                                                        |
| Key Question | Are import partners riskier than export partners?              |
| Chart Type   | HHI Cards + Line Chart + Heatmap + Scatter Plot                |
| KPIs         | Import HHI, Export HHI, Top 3 Import Share, Top 3 Export Share |

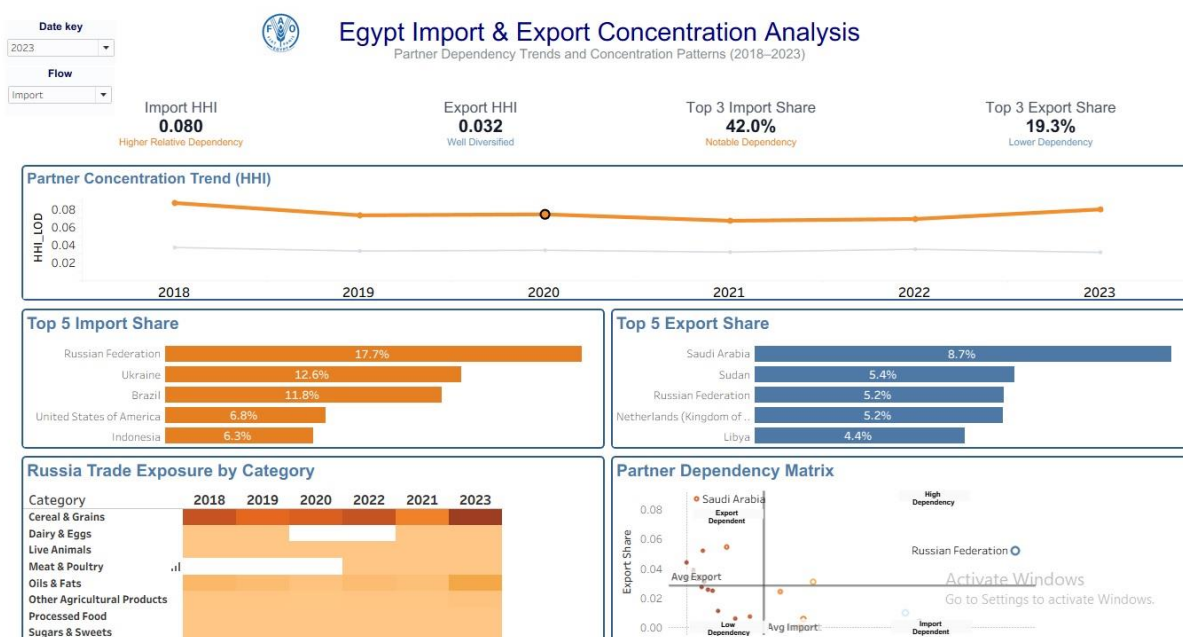
### Results:

| Metric             | Value | Classification               |
|--------------------|-------|------------------------------|
| Import HHI 2023    | 0.080 | ● Higher Relative Dependency |
| Export HHI 2023    | 0.032 | ● Well Diversified           |
| Top 3 Import Share | 42%   | ● Notable Dependency         |
| Top 3 Export Share | 19.3% | ● Lower Dependency           |

### Recommendations:

1. Diversify grain import sources from 3 to 6+ countries (Kazakhstan, Romania, France)
2. Leverage export market diversity (178 countries) for trade negotiations

**Figure 4: Partner Risk & Concentration Dashboard**



### 11.5 Dashboard 5: Growth & Opportunities Matrix

| Element      | Detail                                                             |
|--------------|--------------------------------------------------------------------|
| Tool         | Tableau                                                            |
| Key Question | Which products combine high growth and high stability?             |
| Chart Type   | Bubble Chart (4 Quadrants) + Box Plot                              |
| KPIs         | Highest Growth %, Most Stable (Lowest CV), Stars Count, Average CV |

#### Results:

| Result                 | Value               |
|------------------------|---------------------|
| Highest Growth Product | Chickens (+16,800%) |
| Most Stable Product    | Oranges (CV = 0.12) |
| Stars Count            | 8 products          |
| Average CV             | 0.45                |

#### The 4 Quadrants:

| Quadrant                                                                                   | Meaning                      | Action                |
|--------------------------------------------------------------------------------------------|------------------------------|-----------------------|
| Stars   | High Growth + High Stability | Prioritize investment |
| Bets    | High Growth + Low Stability  | Monitor carefully     |
| Cows    | Low Growth + High Stability  | Improve efficiency    |
| Losers  | Low Growth + Low Stability   | Review or discontinue |

#### Recommendations:

- 1 Prioritize investment in "Stars" quadrant products
- 2 Monitor "Bets" products for sustainable growth
- 3 Improve efficiency for "Cows" products
- 4 Review or discontinue "Losers" products

**Figure 5: Growth & Opportunities Matrix Dashboard**



## 11.6 Dashboard 6: Why Scenario Forecasting? (Intro)

|              |                                                                |
|--------------|----------------------------------------------------------------|
| Element      | Detail                                                         |
| Tool         | Tableau                                                        |
| Key Question | Why is forecasting needed? (Shocks, deficits, deviations)      |
| Chart Type   | Line Chart + Bar Chart                                         |
| KPIs         | Trade Deficit by Year, Coverage Gap to 60%, Trade Shock Impact |

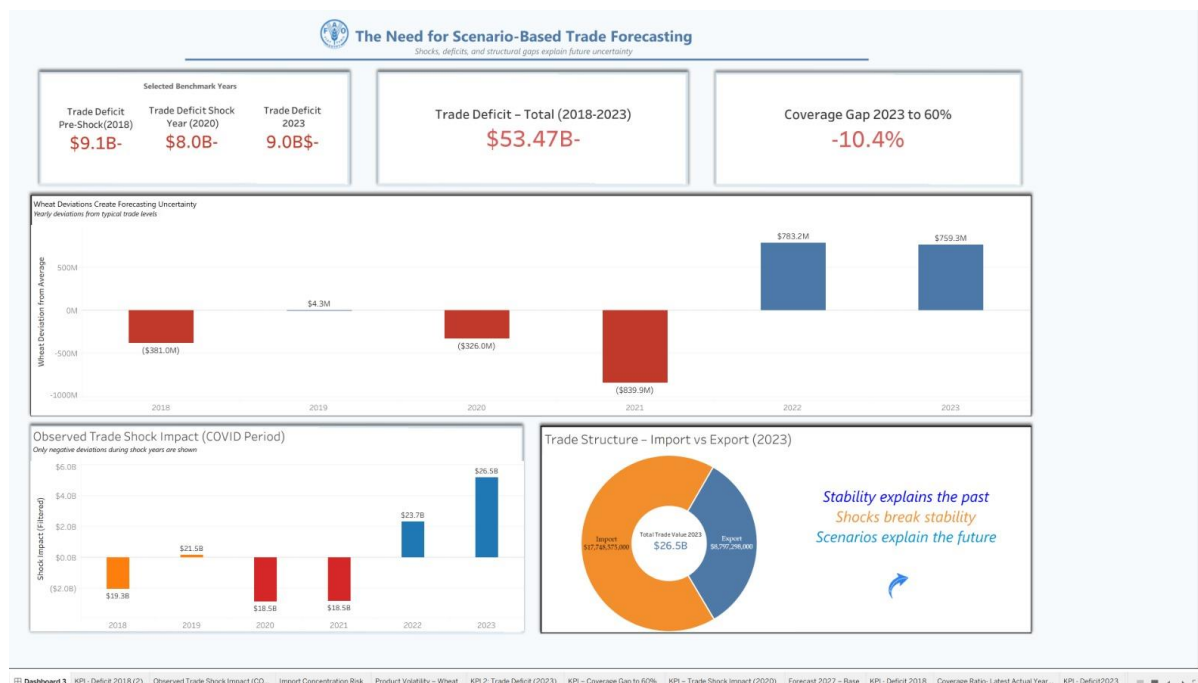
### Key Messages:

- Trade deficit fluctuates with external shocks (COVID, Ukraine war)
- Coverage gap to 60% = additional exports needed
- Shocks break stability → scenarios explain the future

### Recommendations:

- 1 Use scenario planning rather than single-point forecasts
- 2 Prepare contingency plans for pessimistic scenarios

**Figure 6: Why Scenario Forecasting Dashboard**





## 11.7 Dashboard 7: Forecasting & Scenarios

| Element      | Detail                                                        |
|--------------|---------------------------------------------------------------|
| Tool         | Tableau                                                       |
| Key Question | When can coverage ratio reaches 60% and under which scenario? |
| Chart Type   | Line Chart with Forecast + Multiple Scenario Lines            |
| KPIs         | Coverage Ratio 2023 (49.6%), Forecast 2027, MAPE (4.2%)       |

### Results:

| Scenario    | Description                              | Forecast 2027 |
|-------------|------------------------------------------|---------------|
| Optimistic  | Agro-processing + Import diversification | 62-65%        |
| Base Case   | Continuation of current trend            | 58-62%        |
| Pessimistic | Geopolitical or climate shock            | 50-52%        |

### Recommendations:

- 1 Diversify grain import sources to achieve optimistic scenario
- 2 Invest in agro-processing to increase export value

**Figure 7: Forecasting & Scenarios Dashboard**



## 11.8 Dashboard 8: Categories Trends Over Time

|              |                                                                      |
|--------------|----------------------------------------------------------------------|
| Element      | Detail                                                               |
| Tool         | Power BI                                                             |
| Key Question | How does each category's share change over time?                     |
| Chart Type   | Line Chart (Multi-line) + 100% Stacked Bar + Drill through           |
| KPIs         | Total Exports (2023), Total Imports (2023), Fastest Growing Category |

### Results – Import Share Change:

| Category  | 2018 | 2023 | Change                                                                                |
|-----------|------|------|---------------------------------------------------------------------------------------|
| Cereals   | 38%  | 34%  | -4%  |
| Other Ag. | 22%  | 24%  | +2%                                                                                   |
| Meat      | 10%  | 10%  | Stable                                                                                |

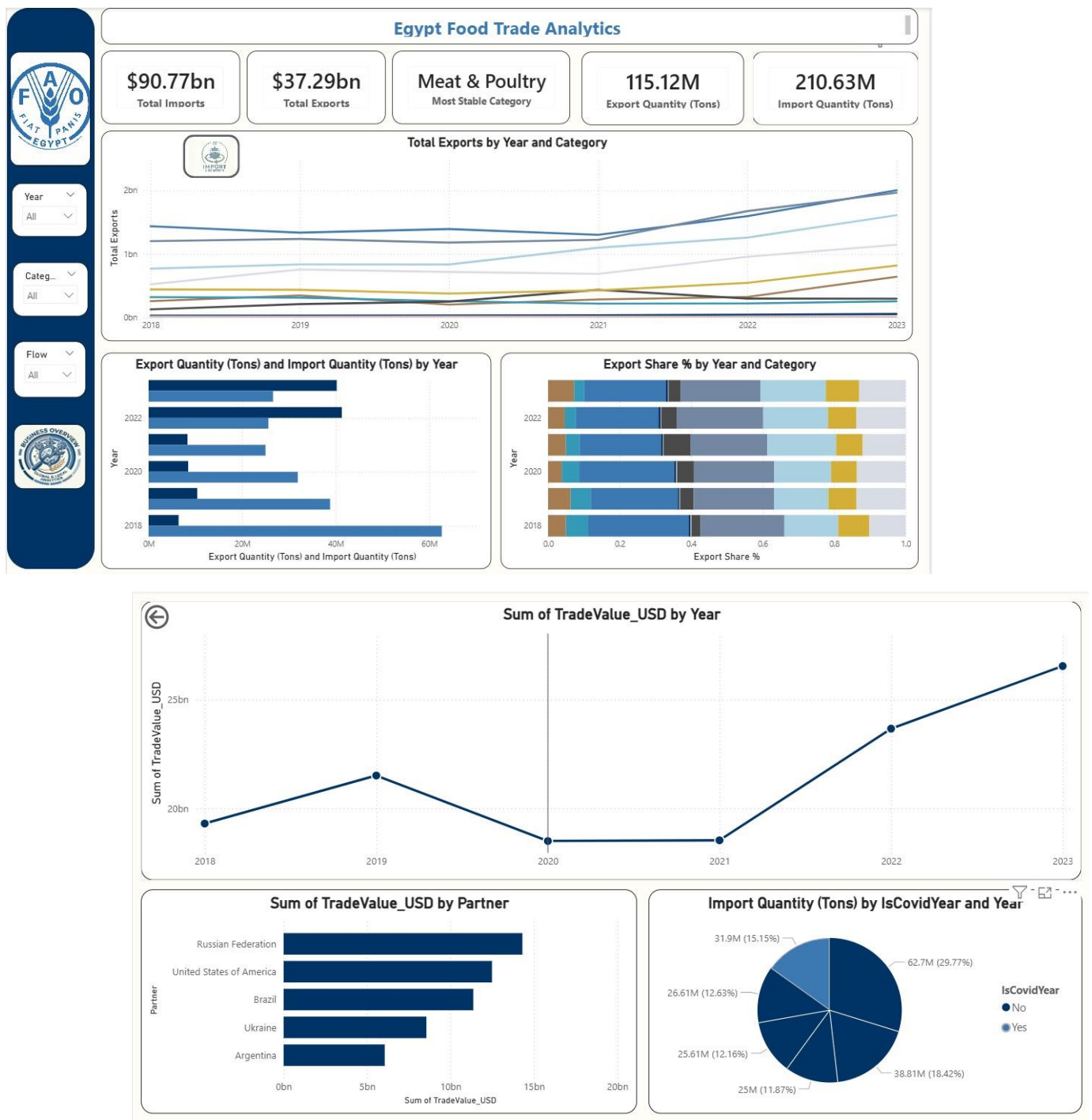
### Results – Export Share Change:

| Category       | 2018 | 2023 | Change                                                                                  |
|----------------|------|------|-----------------------------------------------------------------------------------------|
| Fruits         | 22%  | 24%  | +2%  |
| Processed Food | 16%  | 17%  | +1%                                                                                     |
| Vegetables     | 12%  | 13%  | +1%                                                                                     |

### Recommendations:

- 1 Accelerate export diversification beyond fruits (e.g., processed foods)
- 2 Monitor high-dependency categories quarterly

**Figure 8: Categories Trends Over Time Dashboard**



## 11.9 Dashboard 9: Interactive Executive Dashboard

| Element      | Detail                                                                  |
|--------------|-------------------------------------------------------------------------|
| Tool         | Power BI                                                                |
| Key Question | What is the complete picture and where are risks concentrated?          |
| Chart Type   | Mixed (KPI Cards + Line Chart + Bar Charts + 100% Stacked Bar)          |
| KPIs         | Total Exports, Total Imports, Trade Balance, Coverage Ratio, YoY Growth |

### Main KPIs:

| KPI                       | Value    | Status                                                                              |
|---------------------------|----------|-------------------------------------------------------------------------------------|
| Total Exports (2018-2023) | \$37.3B  |  |
| Total Imports (2018-2023) | \$90.8B  |  |
| Cumulative Trade Balance  | -\$53.5B |  |
| Coverage Ratio 2023       | 49.6%    |  |
| YoY Growth 2023           | +27.1%   |  |

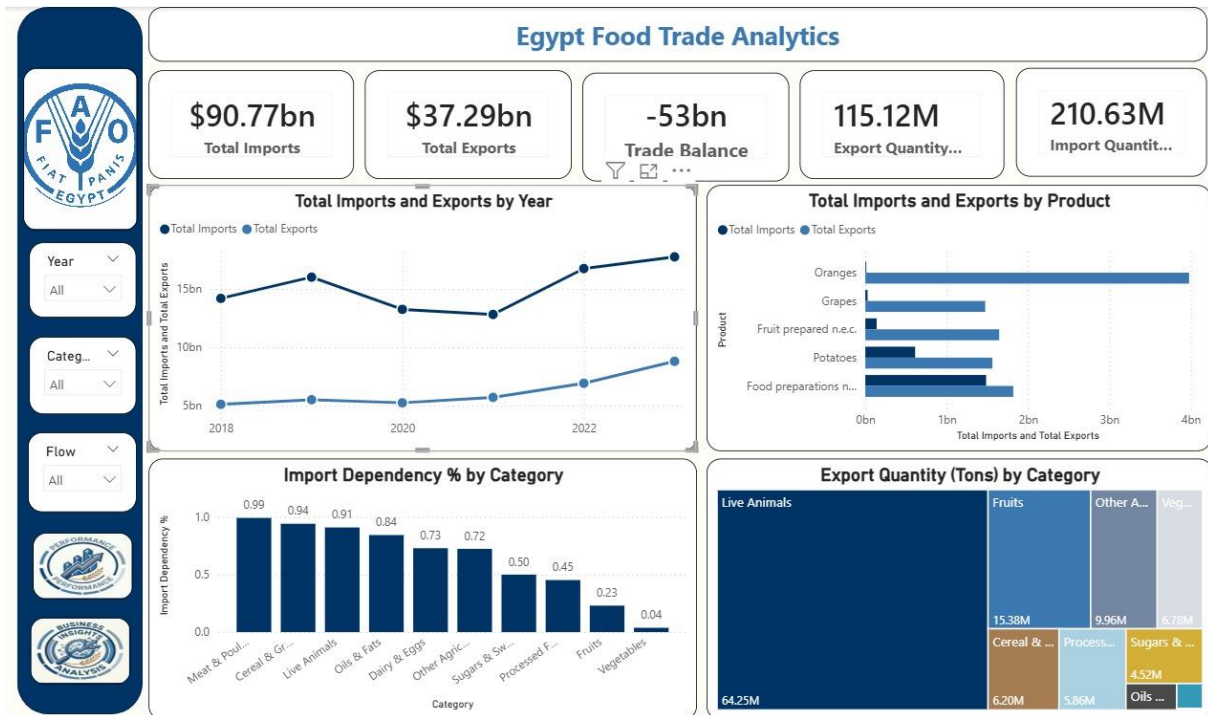
### Risk Summary:

| Risk                                       | Level                                                                                    | Action                                |
|--------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------|
| Import Concentration (3 countries = 39.9%) |  High | Diversify grain import sources        |
| Grain Dependency (94% imported)            |  High | Local production + strategic reserves |
| Meat Gap (99% imported)                    |  High | Livestock investment                  |

### Recommendations:

- 1 Diversify grain import sources from 3 to 6+ countries
- 2 Create monthly monitoring system for key indicators

Figure 9: Interactive Executive Dashboard



## 12. Results & Strategic Recommendations

### 12.1 Key Findings Summary

1. Chronic food trade deficit (-\$53.5B, 2018-2023)
2. Coverage ratio improved from 36% (2018) to 49.6% (2023)
3. 2023 highest export year (\$8.80B) with +27% YoY growth
4. Wheat alone = 14.1% of total trade (\$18.12B)
5. Oranges largest export (\$3.98B) – competitive advantage
6. Meat & Poultry: 99.2% imported – structural gap
7. Cereals & Grains: 94% imported – food security risk
8. Import concentration: Russia + USA + Brazil = 39.9%
9. Export diversification: 178 countries (HHI=0.033)
10. Growth spikes: Chickens (+16,800%), Yoghurt (+10,154%)

### 12.2 Strategic Recommendations

| # | Recommendation                                        | Priority                                                                                       | Responsible Body        |
|---|-------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------|
| 1 | Diversify grain import sources from 3 to 6+ countries |  Urgent       | Ministry of Supply      |
| 2 | Invest in agro-processing for fruits & vegetables     |  High       | Ministry of Agriculture |
| 3 | Develop local meat and poultry production             |  High       | Ministry of Agriculture |
| 4 | Establish strategic grain reserves (6+ months)        |  Medium     | Ministry of Supply      |
| 5 | Create monthly monitoring system for key indicators   |  Continuous | CAPMAS                  |

## 13. User Manual (Dashboard User Guide)

### 13.1 How to Use Tableau Dashboards (Dashboards 4, 5, 6, 7)

| Step | Action                                                      |
|------|-------------------------------------------------------------|
| 1    | Use Year Filter to select a specific year or range          |
| 2    | Hover over any data point for detailed tooltips             |
| 3    | Click on legend items to filter specific categories         |
| 4    | Read the Insight Box at the bottom for key takeaways        |
| 5    | For Dashboard 4: Use Flow Filter (Import/Export) to compare |

### 13.2 How to Use Excel Dashboards (Dashboards 1, 2, 3)

| Step | Action                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Use Slicers (if available) to filter by Year or Category                                                                                                                                                                                                                                              |
| 2    | Follow the color coding:  Good,  Medium,  Critical |
| 3    | Review summary tables for exact numbers                                                                                                                                                                                                                                                               |
| 4    | Check the Insight Box for recommendations                                                                                                                                                                                                                                                             |

### 13.3 How to Use Power BI Dashboards (Dashboards 8, 9)

| Step | Action                                                                                                                                                                                                                                                                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Use Slicers (Year, Category, Flow) to filter data                                                                                                                                                                                                                                                              |
| 2    | Click on any chart element to cross-filter other visuals                                                                                                                                                                                                                                                       |
| 3    | For Dashboard 8: Click on any Category to drill through                                                                                                                                                                                                                                                        |
| 4    | Read the Narrative Visual for automatic insight generation                                                                                                                                                                                                                                                     |
| 5    | Check Risk Flags (    ) for immediate risk assessment |

## 14. References

1. Food and Agriculture Organization of the United Nations (FAO). (2023). *Food and agriculture data (FAOSTAT)*. <https://www.fao.org/faostat>
2. World Bank. (2024). *Egypt economic monitor*. Washington, DC.
3. Central Agency for Public Mobilization and Statistics (CAPMAS). (2024). *Egyptian foreign trade statistics*.

## 15. Appendices

### 1. Appendix A: Raw Dataset Source

FAO Food Trade Dataset (2018–2023)

Source: Food and Agriculture Organization (FAO) Database

File attached separately: raw\_food\_trade\_data.xlsx

### 2. Appendix B: Python Cleaning Code

Separate file attached: data\_cleaning.ipynb

### 3. Appendix C: SQL Scripts & Validation Queries

Separate files attached:

- SQL Schema & ETL Scripts (file: Code.txt)
- Validation Queries (file: تقرير ال سيكوال.docx)

### 4. Appendix D: GitHub Repository

All project source code, data files, dashboards and documentation are available at:

[https://github.com/AminaAdelSaleh/Achievements-Data-Analysis/tree/main/project\\_cleaning\\_phase1](https://github.com/AminaAdelSaleh/Achievements-Data-Analysis/tree/main/project_cleaning_phase1)